

Stationary points



- 1 State the type of point that matches the following descriptions:
 - a A point where the curve changes from decreasing to increasing.
 - b A point where the curve changes from increasing to decreasing.
 - c A point where the tangent is horizontal and the concavity changes about the point.

- 2 For each of the following functions:
 - i Find the derivative.
 - ii Find the coordinates of any stationary points.
 - iii Classify each stationary point.

a $y = -6x^2 + 84x - 29$ c $f(x) = (x + 3)^2(x + 6)$ e $f(x) = -\frac{x^3}{3} + \frac{13x^2}{2} - 30x + 10$ g $f(x) = 134 - 300x + 240x^2 - 64x^3$	b $y = x^3 - 21x^2 + 144x - 19$ d $f(x) = (x + 5)^3 + 4$ f $f(x) = (4x + 5)(x + 1)$ h $f(x) = (x^2 - 9)^2 + 4$
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- 3 Consider the parabola with equation $y = 5 + x - x^2$.
 - a Find the coordinates of the vertex of the parabola.
 - b State the gradient of the tangent to the parabola at the vertex.
 - c What type of stationary point is at the vertex of this parabola?

- 4 Consider the function $f(x) = x^2 + 4x + 9$.
 - a Find an equation for the gradient function $f'(x)$.
 - b State the interval in which the function is increasing.
 - c State the interval in which the function is decreasing.
 - d Find the coordinates of the stationary point.
 - e Classify the stationary point.

5 Consider the function $f(x) = 4x^3 + 5x^2 -$ 

- a Find an equation for the gradient function $f'(x)$.
- b Find the coordinates of the stationary points.
- c Complete the table of values:

x	-2	$-\frac{5}{6}$	$-\frac{1}{2}$	0	1
$f'(x)$		0		0	

d Hence determine the:

- i Local minimum
- ii Local maximum

e Is -4 the absolute minimum value of the function? Explain your answer.

6 Consider the function $f(x) = 3x^2 - 54x + 241$.

- a Find $f'(x)$.
- b Find the x -coordinate of the stationary point.
- c Classify the stationary point.